CLASS 10 — SCIENCE · HARD LEVEL UNIT TEST SET 2

Chapters 9–12: **Light - Reflection and Refraction, The Human Eye and the Colourful World, Electricity, Magnetic Effects of Electric Current** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. Assertion (A): A straight current-carrying wire produces a magnetic field around it. Reason (R): A compass needle placed near the wire gets deflected when the current flows. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

2. A concave mirror forms a virtual image of magnification +2. The object lies at 20 cm. The focal length is 1

(a) -20 cm **(b)** -40 cm **(c)** -10 cm **(d)** -80 cm

3. A person can see only from 40 cm to 100 cm. The lens power for distance and reading lens power (to read at 25 cm) are respectively 1

(a) -2.5 D and +1.5 D **(b)** -1 D and +1.5 D
(c) -1 D and +1 D **(d)** -0.4 D and +3 D

4. A wire is stretched to three times its length with its volume unchanged. The new resistance is 1

(a) R / 3 **(b)** 9 R **(c)** R / 9 **(d)** 3 R

5. A coil moves along the field lines of a uniform field in a direction parallel to the field. The induced current is 1

(a) half **(b)** zero **(c)** maximum **(d)** alternating

6. When the object distance in a convex lens is made 3 times f, the image distance is 1

(a) 2 f **(b)** 1.5 f **(c)** 3 f **(d)** 0.75 f

7. A person who sees the Sun's rays as red at sunset at a place in the plains, will see the rays at the top of a high mountain 1

(a) unchanged
(b) blue
(c) less red than in the plains at the same time
(d) more red

8. In the circuit, ammeter reads 2 A and voltmeter across the resistor reads 10 V. The resistance is 1

(a) 5 ohm **(b)** 20 ohm **(c)** 12 ohm **(d)** 0.2 ohm

9. A magnet is brought toward a coil at a speed v and the induced emf is E. If the speed is tripled, the emf becomes [1]

(a) E **(b)** 3 E **(c)** E / 3 **(d)** 9 E

10. Assertion (A): The image formed by a plane mirror is real. Reason (R): Plane mirror forms an image behind the mirror. [1]

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

11. Assertion (A): Violet light is deviated most by a prism. Reason (R): The refractive index of glass is maximum for violet light. [1]

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

12. Assertion (A): The resistance of a conductor increases with temperature. Reason (R): At a high temperature, the electrons collide more often with the atoms. [1]

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

13. Assertion (A): Soft iron is used as the core of an electromagnet. Reason (R): Soft iron is easily magnetised and demagnetised. [1]

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

14. For an object between P and F of a concave mirror, as the object moves from F towards P, the image size [1]

(a) decreases **(b)** becomes zero first
(c) stays the same **(d)** increases

15. The ability of the eye to see objects clearly at different distances depends on [1]

(a) the retina moving back and forth
(b) the pupil moving the lens
(c) the ciliary muscles changing the shape of the lens
(d) the cornea changing its colour

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. If the angle of the incidence on the prism is varied, will the colours separate in the same way? State why. [2]

17. A heater of 800 W is connected to a 220 V source. Find the resistance and the heat produced in 5 minutes. [2]

18. Calculate the current drawn by a 2.2 kW geyser at 220 V and say whether a 5 A or a 15 A circuit is suitable. 2

19. Find the lateral shift reasoning: why is it zero for normal incidence? 2

20. How does the iris protect the eye and how does the lens focus light? 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. Why should an electrical appliance such as a toaster have a metal casing connected to the earth wire? 3

22. A ray of light enters from glass (n = 1.5) to water (n = 1.33). Does it bend towards the normal? Find the relative refractive index. 3

23. An object is placed at 10 cm and 100 cm in front of a normal eye. At which position does the ciliary muscle contract more and why? 3

24. Why are ammeters of low resistance and voltmeters of high resistance? 3

OR

Why do we say that electricity and magnetism are related? Give two examples. 3

25. A convex lens forms a real image of a candle on a screen 80 cm away. The magnification is -3. Find u and f. 3

OR

An elderly person has a near point of 1 m and a far point of 2 m. Describe the defects and the lens prescription for reading at 25 cm and seeing far. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. A point source is placed 30 cm from a concave mirror of f = 20 cm. By how much and in which direction should the source move so that the image shifts to infinity? Explain what the beam becomes. 5

OR

An eye of a person has near point at 25 cm and far point at 2 m. Find the power of lens that can bring distant objects into clear focus. Predict the nature of the image formed by the lens of an object at infinity. 5

27. An electric heater has two coils of 20 ohm and 30 ohm that can be used singly, in series or in parallel on a 120 V supply. Find the power in each of the four modes and identify the highest-power mode. 5

OR

Compare the magnetic field of a bar magnet and of a current-carrying solenoid. Explain how a solenoid can be used to lift iron scrap. 5

END OF QUESTION PAPER — Total: 50 Marks